

Use the given conditions to write an equation for the line in slope-intercept form, if possible.

1. Slope = -3 , passing through $(2, 6)$.

2. The line is parallel to $y = -\frac{1}{2}x + 5$ and passes through $(2, 0)$.

3. Through $(5, 0)$ and $(-6, 2)$.

4. Through $(1, 2)$ and $(3, 2)$.

5. Through $(3, 1)$ and $(3, 2)$.

Solve the problem.

6. After 5 years as an actuary, Sue's salary was \$130,000. After 12 years her salary had risen to \$200,000. Assuming that the change in her salary over time is linear, find an equation expressing her salary as y and x as the number of years on the job.
7. Using the equation you found in 6, how many years does she need to work as an actuary to have a salary of \$250,000?
8. A shoe company will make a new type of shoe which costs \$21 to make per shoe. The fixed cost of producing this shoe (utilities, property taxes, leases, etc.) is \$25,000. The company plans to sell the shoe for \$100 a pair.
- Find the cost function in dollars, $C(x)$, for x pairs of shoes.
 - Find the revenue function in dollars, $R(x)$, for x pairs of shoes.
 - Find the profit function in dollars, $P(x)$, for x pairs of shoes.
 - How many pair of shoes must they sell to break even?